

Exploring multi-model orchestration for high-frequency trading platforms

Daniel-Ştefan Halasz
vu\dha242
Student no. 2804542
d.halasz@student.vu.nl

Vitor Andrade Fernández
vu\van204
Student no. 2798043
v.andrade.fernandez@student.vu.nl

Karina Kalicka-Molin
vu\qjy548
Student no. 2861685
k.b.kalicka-molin@student.vu.nl

Abstract—Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum. Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus.

I. INTRODUCTION

PREDICTING market trends is a critical aspect of making informed trading decisions in the financial industry. Accurate predictions lead to more profitable investment strategies and increased returns. Symmetrically, inaccurate predictions carry significant risks, ranging from financial losses to missed opportunities. Machine learning (ML) techniques have been increasingly applied to the finance sector, aiming to increase the accuracy of market predictions. Kelly et al. [1] provide an overview of the shift towards data-driven architectures in financial modeling, highlighting the potential of ML to enhance predictive capabilities. In short, the finance industry has witnessed a significant paradigm shift in its approach to estimating market behavior, transitioning from traditional linear models to complex ML-based models that can capture non-linear interactions in financial data. With the newly perfected ML-based strategies, investors can execute trades at unprecedented speeds, capitalizing on even the smallest market movements; thus, high-frequency trading (HFT) has become a dominant force in the financial markets. As of late 2025, empirical evidence indicates that the footprint of HFT in the US financial markets reached approximately 73% of the total trading volume, while in the European markets, HFT accounts for around 50% of the trading volume [2]–[4]. Therefore,

work towards improving the accuracy of HFT strategies, while reducing model inference is highly rewardable, as it can yield increased financial gains for investors and firms operating in the financial markets.

The adoption of ML-based strategies in trading poses significant challenges, particularly in the context of HFT (or in other words, short-term trading), where the market is unpredictable in its volatility, and highly dynamic in its behavior. In such conditions, a reoccurring problem is the lack of a generalizable model that can consistently outperform the market across different time periods and market conditions. For example, research shows that recurrent neural network (RNN)-based models excel in capturing complex, non-linear relationships in high-volatility market regimes, while linear models perform better during stable periods, because they are more likely to overfit on noise data, which is a characteristic of fast-moving markets [5], [6]. This phenomenon (often referred to as the *no free lunch* problem) highlights the need for adaptive strategies in HFT platforms. Current work [4] towards addressing the need of adaptability disregards the computational latency cost of the proposed solutions, which is a critical aspect in HFT platforms. If the latency of the prediction model exceeds the shelf-life of the predictive signal, the accuracy gain becomes economically irrelevant. A review of current literature reveals a gap in research that explicitly considers the trade-off between prediction accuracy and computational latency in adaptive market prediction strategies.

In this work, we propose a novel approach towards adaptive market prediction in HFT platforms, which is based on an orchestrator model that dynamically selects the most suitable prediction model based on current market conditions. The introduction of an orchestrator introduces an additional layer of complexity, and thus, increased latency, which can potentially negate the benefits of improved prediction accuracy. We address the following main research question: **How to maximize the asset gains of HFT platforms using an orchestrated multi-model prediction approach?** We analyze the accuracy-latency trade-off in the context of several orchestrator designs, and experimentally evaluate their performance using real-world financial data. Furthermore, we compare the performance of the orchestrated approach against traditional single-model strategies, and analyze the conditions under which an orchestrator provides a significant advantage.

II. BACKGROUND

III. METHODOLOGY

In this section, we analyze the design choices and trade-offs involved in an orchestrator-based approach to adaptive market prediction models for HFT applications. Our approach is structured in three stages: (i) the analysis of data and features involved in market prediction models, (ii) the design of orchestrator architectures based on the aggregate performance of a set of baseline models, and (iii) the evaluation of orchestrator performance against the baselines using a comprehensive set of metrics.

Note that this work is not focused on designing a new baseline market prediction model, but rather use existing models as a basis for the orchestrator design and evaluation. To answer the main research question, we focus on the design of a meta-model that can select the best-performing baseline model for a given market regime, based on a set of meta-features that capture both the current market state and the recent performance of the baseline models.

A. Datasets & feature space

All market data is sourced from Yahoo Finance via the `yfinance` Python package [?], accessed at two resolutions. For base-model training the *9000+ Tickers of Stock Market Data (Full History)* dataset [?] is used. It is a pre-packaged collection of daily OHLCV records spanning 1962-2024 across more than 9000 US equity tickers (≈ 3.5 GB), a breadth that ensures that models are exposed to a wide variety of market regimes, sectors, and volatility environments. For latency benchmarking and higher-frequency validation we query the same Yahoo Finance source directly for 5-minute intraday bars over the most recent 60 days, keeping data provenance consistent across both resolutions.

In order to avoid look-ahead bias, all splits are strictly chronological: daily data from 2015-2022 is used for training the base models, and 2023 serves as the validation set from which orchestrator training labels are derived (see Section III-D), and 2024 is held exclusively for final evaluation. No shuffling is applied at any stage.

Raw OHLCV data are not sufficient for short-horizon prediction; therefore engineer features come from three groups grounded in the LOB microstructure literature [?], [?]. *Price microstructure* features include the mid-price $p_{\text{mid}} = (p_{\text{high}} + p_{\text{low}})/2$, the spread proxy $s_t = p_{\text{high},t} - p_{\text{low},t}$ (a liquidity indicator), log return $r_t = \ln(p_t/p_{t-1})$, and rolling realized volatility $\hat{\sigma}_t$.

Order-flow features include the volume imbalance ratio as an OFI proxy [?]:

$$\text{OFI}_t = \frac{V_t - V_{t-1}}{V_t + V_{t-1} + \varepsilon}, \quad (1)$$

and VWAP deviation $\delta_t = p_t - \text{VWAP}_t$, capturing the mean-reversion pressure.

Technical indicators (RSI-14, EMA-9, EMA-21, MACD, Bollinger Band width) complement the microstructure signal.

The resulting feature vector per bar is

$$\mathbf{x}_t = \begin{bmatrix} r_t, \hat{\sigma}_t, s_t, \text{OFI}_t, \delta_t, \text{RSI}_{14,t}, \text{EMA}_{9,t}, \\ \text{EMA}_{21,t}, \text{MACD}_t, \text{BB width}_t \end{bmatrix}^\top \in \mathbb{R}^{10}. \quad (2)$$

B. Baseline market prediction models

C. Evaluation metrics

D. Feature Engineering for Model Orchestration

Rather than operating on raw market data, the orchestrator takes as input a *meta-feature vector* $\mathbf{z}_t$ built from three groups of signals:

$$\mathbf{z}_t = \left[\underbrace{s_t, \hat{\sigma}_t, \text{OFI}_t}_{\text{regime}} \mid \underbrace{\hat{y}_t^{(1)}, \hat{y}_t^{(2)}, \hat{y}_t^{(3)}}_{\text{base outputs}} \mid \underbrace{\ell_{t-\tau}^{(1)}, \ell_{t-\tau}^{(2)}, \ell_{t-\tau}^{(3)}}_{\text{rolling loss}} \right]^\top, \quad (5)$$

where $\hat{y}_t^{(k)}$ is model k 's predicted upward-movement probability and $\ell_{t-\tau}^{(k)}$ is its mean cross-entropy loss over the preceding $\tau = 20$ bars. The regime indicators capture the current market state, which strongly influences which model performs best [?].

Training Labels: To generate training labels, all three base models are run in a forward pass over the validation split (2023), keeping the orchestrator's training data separate from the base models'. The ground-truth label k_t^* , hereafter referred to as the *oracle label*, is defined as the base model with the lowest rolling prediction error over the preceding τ bars:

$$k_t^* = \arg \min_k \frac{1}{\tau} \sum_{i=0}^{\tau-1} \mathbf{1}[\hat{y}_{t-i}^{(k)} \neq y_{t-i}]. \quad (6)$$

This frames orchestration as a 3-class classification task, trained with categorical cross-entropy:

$$\mathcal{L}_{\text{orch}} = -\frac{1}{T} \sum_{t=1}^T \sum_{k=1}^3 \mathbf{1}[k_t^* = k] \log \hat{g}_t^{(k)}, \quad (7)$$

where $\hat{g}_t^{(k)}$ is the gating network's softmax weight for model k . Class imbalance is corrected by inverse-frequency weighting, missing values are filled by linear interpolation, and all features in $\mathbf{z}_t$ are standardized on the training split only.

E. Orchestrator market prediction meta-model architecture

F. Orchestrator evaluation & trade-offs

IV. EXPERIMENTAL SETUP

V. RESULTS

VI. DISCUSSION & RESULT ANALYSIS

VII. FUTURE WORK

VIII. CONCLUSION

REFERENCES

- [1] S. Gu, B. Kelly, and D. Xiu, "Empirical asset pricing via machine learning," *The Review of Financial Studies*, vol. 33, no. 5, pp. 2223–2273, 2020.

- [2] R. Abdelghani, "Evaluation of high-frequency prediction approaches in price determination on financial markets," *European Public & Social Innovation Review*, vol. 10, pp. 1–15, 01 2025.
- [3] K. Batoool, M. M. Baig, and U. Fatima, "Meta-learning for financial market prediction: An efficient approach with reduced computational cost," *International Journal of Advanced and Applied Sciences*, vol. 12, no. 11, pp. 72–81, 2025.
- [4] K. Noor and U. Fatima, "Meta learning strategies for comparative and efficient adaptation for financial datasets," *IEEE Access / ResearchGate Preprint*, vol. 13, p. 24159, 2025. [Online]. Available: <https://www.researchgate.net/publication/388914837>
- [5] R. Contributors, "Adaptive market intelligence: A mixture of experts framework for volatility-sensitive stock forecasting," *ResearchGate / International Journal of Forecasting*, 2025. [Online]. Available: <https://www.researchgate.net/publication/394322530>
- [6] A. W. Lo, "The adaptive market hypothesis," *Journal of Portfolio Management*, vol. 30, no. 5, pp. 15–29, 2004.

APPENDIX

LIST OF ACRONYMS

HFT high-frequency trading
ML machine learning
RNN recurrent neural network